TANVIR HASAN SHOVON

+358417216414⋄ Tampere, Finland
tanvir.shovon@tuni.fi ⋄ LinkedIn ⋄ GitHub

EDUCATION

M.Sc, Tampere University, Finland 2023 - 2025
Area: signal processing and Machine learning
Keywords: Machine learning, Computer vision, Signal processing, Image processing, Audio processing
Thesis: Hybrid ViViT–TimeSformer Transformer Architecture for Multimodal Slip Detection in Robotic Manipulation (Grade: 5)
GPA: 4.52 out of 5.00

B.Sc, Khulna University of Engineering Technology, Bangladesh 2015 - 2019
Area: Electronics and Communication Engineering
Thesis: Classification of EEG Signals with multi-input Convolutional Neural Network by augmenting STFT
GPA: 3.62 out of 4.00

EXPERIENCE

FAST-Lab, Tampere University March 2025 - Present
Research Assistant *Tampere, Finland*

- Conducting research at the intersection of AI, robotics, and machine learning under Professor Jose Martinez Lastra, focused on transformer-based models for deformable object manipulation to enhance robotic automation in smart manufacturing.
- Designed a dual-backbone TimeSformer-ViViT architecture for multimodal visuo-tactile slip detection, fusing RGB and tactile sensor streams.

Computer Vision Research Group, Tampere University January 2025 - February 2025
Teaching Assistant *Tampere, Finland*

- Directed exercise sessions and solutions. Supervised exams and coordinated other teaching assistants.

Audio Research Group, Tampere University October 2024 -December 2024
Teaching Assistant *Tampere, Finland*

- Directed exercise sessions and solutions. Supervised exams and coordinated other teaching assistants.

Computer Vision Group, Tampere University June 2024 -August 2024
Research Assistant *Tampere, Finland*

- Developed and implemented a real-time multi-camera 6D pose estimation and localization system for accurately tracking object movements and UAVs in dynamic environments using advanced machine learning techniques under Professor Joni Kämäräinen. .

Signal Processing and Machine Learning Group, KUET August 2021 -2022
Research Assistant *Khulna, Bangladesh*

- Designed and implemented deep learning pipelines for medical image and biosignal analysis: VGG16 transfer learning for breast ultrasound cancer classification, and a multi-input CNN with STFT-based EEG augmentation for motor-imagery classification.

PUBLICATIONS

T.H. Shovon, Z.A. Nazi, S. Dash and M.F. Hossain, Classification of Motor Imagery EEG Signals with multi-input Convolutional Neural Network by augmenting STFT, *2019 5th International Conference on Advances in Electrical Engineering (ICAEE)*, 2019, pp. 398-403, doi:10.1109/ICAEE48663.2019.8975578 Link

Under review: Transformer-based multimodal visuo-tactile slip detection for robotic manipulation (journal). Link

In preparation: Sensitivity-aware quantization budget reallocation for sparse-quantized transformers.

In preparation: Energy-guided layer localization for sparse autoencoder-based faithfulness detection.

PROJECTS

- **Sensitivity-Aware Quantization Budget Allocation:** Reallocating base precision, group size and outlier rate across layers under a fixed average bit-width, on the finding that weight sensitivity varies sharply with layer depth and role. Evaluated on LLaMA-2/3, Mistral and OPT.
- **Attention Efficiency Study:** Systematic comparison of GQA, MQA, MLA and TPA with post-training quantization benchmarking across small open models (Qwen, Llama).
- **Diffusion-Based Real Image Editing:** Null-text inversion with Delta Denoising Score guidance for text-driven editing of real images, studying how inversion fidelity constrains edit quality. Link
- **WinterNav-RUGD:** Safety-oriented traversability mapping for small autonomous ground vehicles. Link

HONOURS & AWARDS

Tampere University scholarship 2023-2025

Dean's Award 2019

Won the **prize** from the Faculty of Electrical and Electronic Engineering (EEE), KUET.

KUET University scholarship 2016-2018

TECHNICAL SKILLS

DeepNN	PyTorch, TensorFlow, Keras
Languages and Tech	Python, Matlab, c++,Bash Scripting, Linux
Softwares	OpenCV, Jupyter Notebook, Scikit-learn, LATEX, Qt Creator, LATEX
Version Control & CI/CD	Git, GitHub, Docker-Compose

COURSES

- Artificial Intelligence, DATA.ML.310
- Computer Vision, DATA.ML.300
- Media Analysis, DATA.ML.330
- Introduction to Audio Processing, COMP.SGN.120
- Introduction to Image and Video Processing, COMP.SGN.110
- Introduction to Signal Processing, COMP.SGN.100
- Introduction to Pattern Recognition and Machine Learning, DATA.ML.100
- COMP.CS.300 Data Structures and Algorithms 1
- COMP.CS.500 Introduction to Full Stack Development

REFERENCES

- **Joni Kämäräinen**, Professor of Signal Processing
  joni.kamarainen@tuni.fi
  Computer Vision Group, Tampere University
- **Jose Martinez Lastra**, Professor of Factory Automation
  jose.martinezlastra@tuni.fi
  Faculty of Engineering and Natural Sciences, Automation Technology and Mechanical Engineering Tampere University